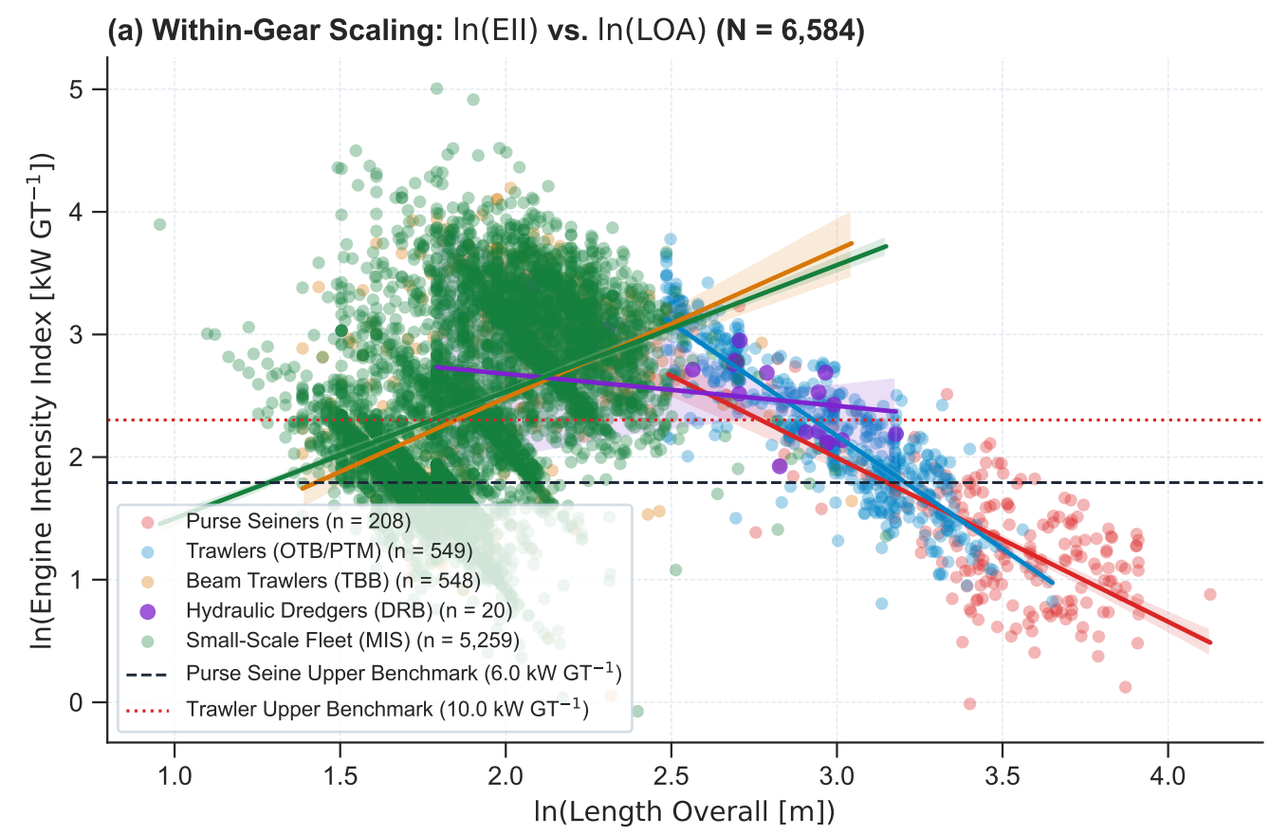
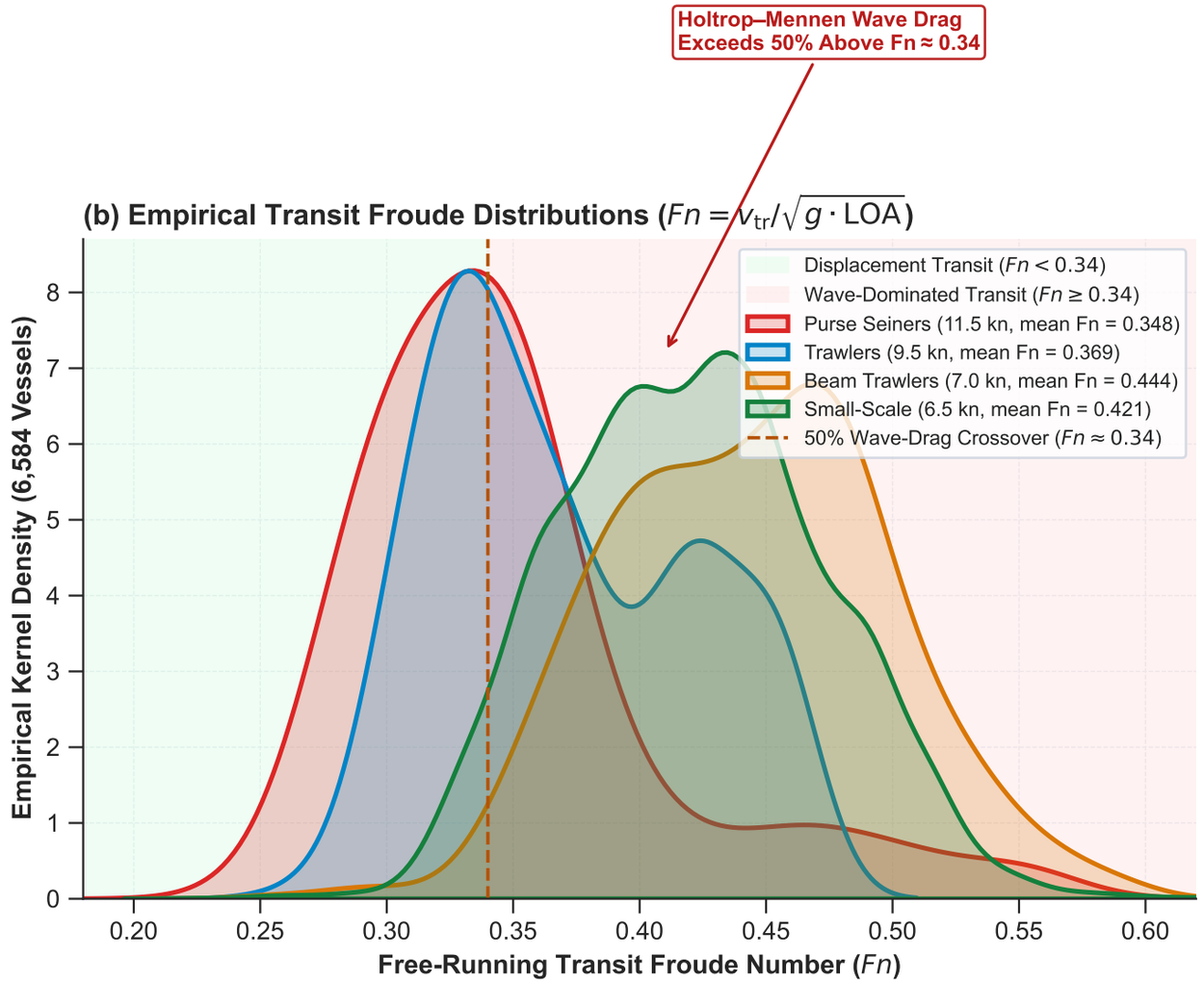
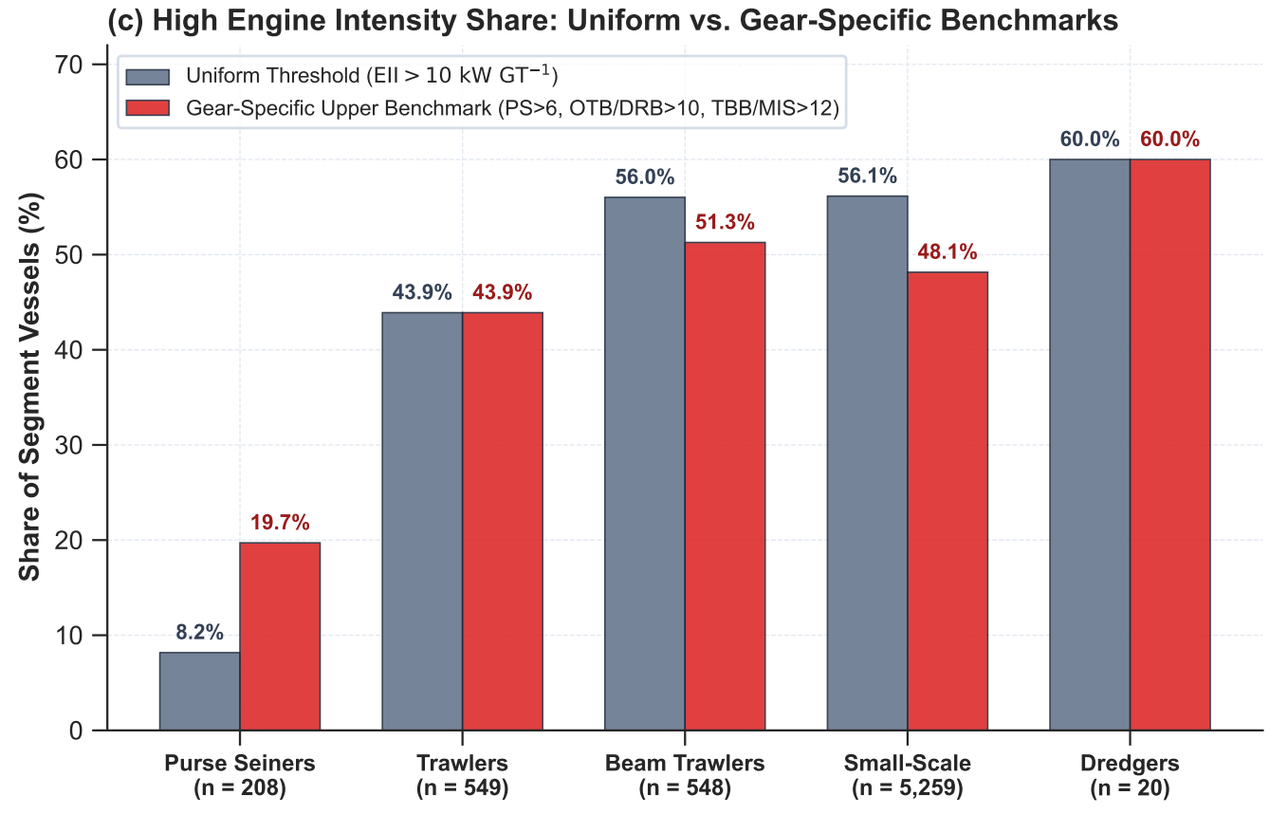
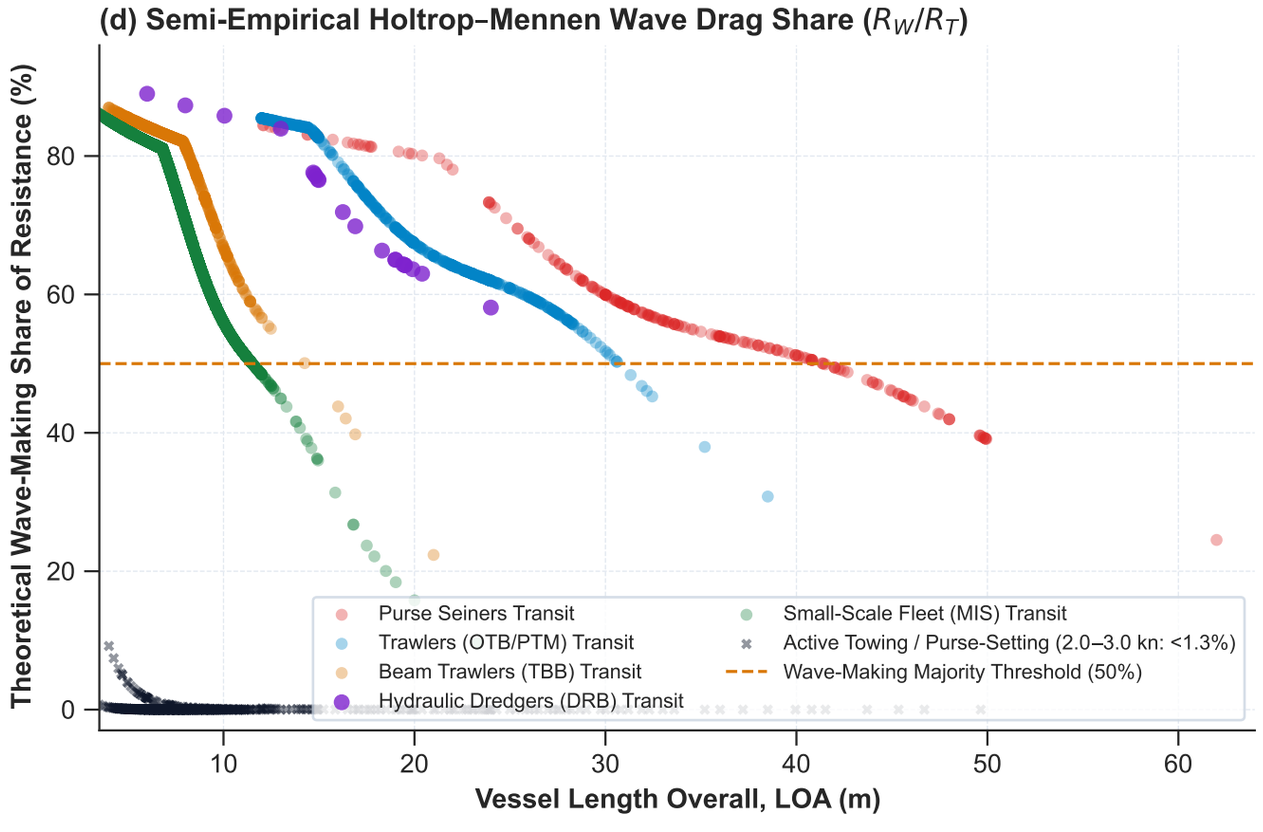

(a) Within-Gear Scaling: ln(EII) vs. ln(LOA) (N = 6,584)
ln(Engine Intensity Index [kW GT⁻¹])
ln(Length Overall [m])
Purse Seiners (n = 208)
Trawlers (OTB/PTM) (n = 549)
Beam Trawlers (TBB) (n = 548)
Hydraulic Dredgers (DRB) (n = 20)
Small-Scale Fleet (MIS) (n = 5,259)
Purse Seine Upper Benchmark (6.0 kW GT⁻¹)
Trawler Upper Benchmark (10.0 kW GT⁻¹)


Holtrop–Mennen Wave Drag
Exceeds 50% Above Fn ≈ 0.34
(b) Empirical Transit Froude Distributions (Fn = v_tr/√(g · LOA))
Empirical Kernel Density (6,584 Vessels)
Free-Running Transit Froude Number (Fn)
Displacement Transit (Fn < 0.34)
Wave-Dominated Transit (Fn ≥ 0.34)
Purse Seiners (11.5 kn, mean Fn = 0.348)
Trawlers (9.5 kn, mean Fn = 0.369)
Beam Trawlers (7.0 kn, mean Fn = 0.444)
Small-Scale (6.5 kn, mean Fn = 0.421)
50% Wave-Drag Crossover (Fn ≈ 0.34)


(c) High Engine Intensity Share: Uniform vs. Gear-Specific Benchmarks
Share of Segment Vessels (%)
Uniform Threshold (EII > 10 kW GT⁻¹)
Gear-Specific Upper Benchmark (PS>6, OTB/DRB>10, TBB/MIS>12)
8.2%
19.7%
43.9%
43.9%
56.0%
51.3%
56.1%
48.1%
60.0%
60.0%
Purse Seiners (n = 208)
Trawlers (n = 549)
Beam Trawlers (n = 548)
Small-Scale (n = 5,259)
Dredgers (n = 20)


(d) Semi-Empirical Holtrop–Mennen Wave Drag Share (R_W/R_T)
Theoretical Wave-Making Share of Resistance (%)
Vessel Length Overall, LOA (m)
Purse Seiners Transit
Trawlers (OTB/PTM) Transit
Beam Trawlers (TBB) Transit
Hydraulic Dredgers (DRB) Transit
Small-Scale Fleet (MIS) Transit
Active Towing / Purse-Setting (2.0–3.0 kn: <1.3%)
Wave-Making Majority Threshold (50%)